

Challenges in Training PINNs: A Loss Landscape Perspective

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Why PINNs matter and why they struggle

- Partial differential equations (PDEs) govern fluid flow, waves, heat, and electromagnetism across science and engineering.
- Physics-Informed Neural Networks (PINNs) train a neural function to satisfy the PDE and the boundary and initial conditions.
- On challenging problems, training often stalls at high loss due to severe ill-conditioning induced by differential operators.
- We analyze the loss landscape and present an optimization recipe that reliably drives the loss near zero and improves accuracy.

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What is optimized in a PINN

- Training objective: total loss = PDE residual term + boundary-condition term + initial-condition term.
- We study behavior across representative convection, reaction, and wave problems to cover different operator types.

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Loss-landscape evidence: ill-conditioning

- Hessian spectra show large outlier eigenvalues and heavy mass near zero, implying very high condition numbers.
- The PDE residual term drives most of the ill-conditioning; limited-memory BFGS (L-BFGS) substantially tightens the spectrum.

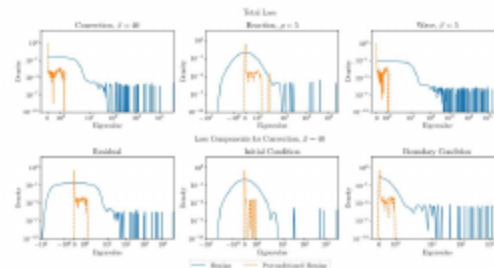


Figure: Notice the spectrum's large outliers and mass near zero; after quasi-Newton updates, the spectrum becomes significantly tighter.

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- Lower final loss consistently yields lower relative L2 error (L2RE).
- Accurate solutions usually require driving the loss very close to zero.

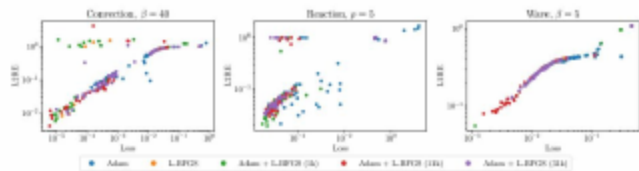


Figure: Notice the strong correlation: models with lower training loss achieve lower relative L2 error across PDEs and widths.

- Stage 1 (Adam): explores the landscape and escapes saddles.
- Stage 2 (L-BFGS): exploits curvature for fast local descent; a brief Newton-type finetune can make further progress if L-BFGS plateaus.

(In the plot, the final stage is labeled NNCG, i.e., our Newton-type finetuner.)

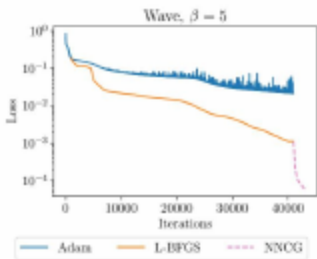


Figure: Notice the training curves: Adam plateaus on difficult problems, L-BFGS lowers the loss, and a brief Newton-type finetune drives it further down.

- A Newton-type finetuner reliably cuts the loss by over 10 $\times$ after Adam+L-BFGS, while gradient-descent finetuning fails in the same regime.
- Use it briefly at the end due to higher per-iteration cost.

(Plots label NNCG for the Newton-type finetuner and GD for gradient descent.)

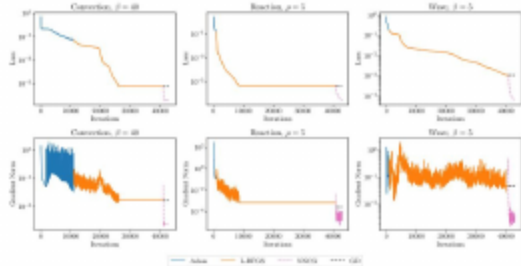


Figure: Notice the gap: the Newton-type finetune continues reducing loss and gradients where gradient-only finetuning makes no progress.

- Error fields shrink across stages: Adam $\rightarrow$ L-BFGS $\rightarrow$ Newton-type finetune.
- Notice structured streaks left after L-BFGS that the finetune removes, yielding smoother, more accurate fields.

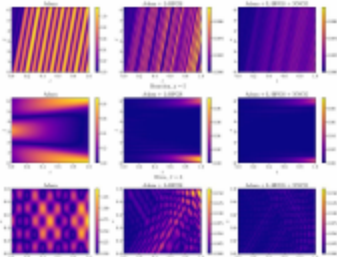


Figure: Notice the progression: left (Adam) shows structured errors, middle (L-BFGS) reduces them, right (Newton-type) removes remaining artifacts.

- Low PDE residual alone can be misleading if boundary or initial conditions are unmet—leading to large solution error; balance the loss terms.
- Practical recipe: Adam → L-BFGS → brief Newton-type finetune.

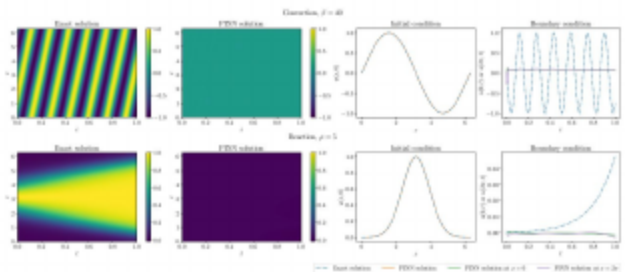


Figure: Notice the failure case: a near-constant solution fits the residual but violates conditions, producing high overall error.

- Questions and feedback are welcome.